

Familial Neurodevelopmental Disorder with Developmental Delay and Strabismus

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Abstract

Background: Neurodevelopmental disorders (NDDs) are genetically heterogeneous, and a substantial fraction of cases still lack a molecular diagnosis. CX3CR1 encodes the fractalkine receptor and is central to neuron–microglia communication, microglial homeostasis, and synaptic remodeling during brain development. Although common CX3CR1 variants have been studied extensively in neurological and inflammatory conditions, their possible contribution to familial neurodevelopmental phenotypes has received little attention.

Case Presentation: We describe a male proband from a non-consanguineous family who presented with developmental delay, intellectual disability, and strabismus. Three of six siblings were affected (two females and one male), and an affected maternal male first cousin was also reported. The pedigree is compatible with autosomal recessive inheritance, yet this could not be confirmed because only the proband underwent molecular testing and segregation analysis was unavailable.

Genetic Findings: Whole-exome sequencing (WES) of the proband was performed. After filtering out benign/likely benign variants and intronic changes, and after reviewing genes previously linked to the phenotype, no novel or clearly pathogenic variants remained. Two heterozygous missense variants in CX3CR1—c.745G>A (p.Val249Ile; V249I; rs3732379) and c.839C>T (p.Thr280Met; T280M; rs3732378)—were retained as candidates. Both are well-known common functional polymorphisms that frequently travel together in linkage disequilibrium. Their heterozygous state alone does not prove pathogenicity, and phase (cis versus trans) as well as familial segregation could not be determined.

Conclusion: The presence of these two CX3CR1 variants in a family with a clear neurodevelopmental phenotype, together with the established role of CX3CR1 in microglial and neuronal biology, makes CX3CR1 a reasonable candidate gene for further study. Current evidence, however, falls short of establishing causality. Parental testing, segregation analysis, molecular phasing, functional assays, and additional independent cases will be required to test this hypothesis properly.

Keywords: *CX3CR1; neurodevelopmental disorder; developmental delay; intellectual disability; strabismus; whole-exome sequencing; candidate gene; neuron–microglia signaling.*

1. Introduction

Neurodevelopmental disorders form a broad and genetically diverse group of conditions that impair cognition, behavior, and adaptive skills. Despite major advances in sequencing, a large proportion of individuals still receive no molecular diagnosis. Exome sequencing has become a first-line diagnostic tool for unexplained developmental delay and intellectual disability because it interrogates virtually all protein-coding genes in a single assay and yields a diagnosis in a substantial fraction of patients [1].

Neuron–microglia interactions are increasingly recognized as essential for normal brain development and synaptic maturation. CX3CR1 encodes the sole receptor for the chemokine CX3CL1 (fractalkine). In the central nervous system this receptor is expressed almost exclusively on microglia and mediates bidirectional signaling with CX3CL1-expressing neurons [2,3]. Mouse studies have shown that loss of *Cx3cr1* produces transient changes in microglial density, defective synaptic pruning, reduced synaptic transmission, weakened functional connectivity, and behavioral phenotypes relevant to neurodevelopmental and neuropsychiatric disorders [3]. These experimental data supply a clear biological rationale for considering CX3CR1 in human neurodevelopmental phenotypes.

Human genetic data are still limited but suggestive. Ishizuka and colleagues resequenced CX3CR1 in cohorts with schizophrenia and autism spectrum disorder and identified a rare p.Ala55Thr variant that associated with both phenotypes. Structural modeling and functional assays indicated that the substitution impairs fractalkine-

dependent signaling [4]. Although this rare allele is distinct from the common polymorphisms discussed below, the finding strengthens the case for examining CX3CR1 variation in neurodevelopmental contexts.

Two common nonsynonymous variants, V249I (rs3732379) and T280M (rs3732378), have been studied far more extensively. They lie in the sixth and seventh transmembrane domains, respectively, and have been shown to alter receptor function in several cellular assays [5,6]. In most populations the two sites are in near-complete linkage disequilibrium, so the I249/M280 haplotype behaves as a single allele [7]. Reported associations span Parkinson's disease, multiple sclerosis, amyotrophic lateral sclerosis, and various inflammatory or vascular traits, yet results have been inconsistent across cohorts [8–10]. Collectively these data support the view that V249I and T280M are functional polymorphisms rather than high-penetrance Mendelian alleles.

Here we report whole-exome sequencing of a male proband from a non-consanguineous family with developmental delay, intellectual disability, and strabismus. After standard filtering, two heterozygous CX3CR1 variants—V249I and T280M—remained as the only candidates of interest. No potentially causative variants were identified in genes previously linked to the phenotype. Given the biological role of CX3CR1 and the prior human genetic hints, we put CX3CR1 forward as a candidate gene. At the same time we emphasize that the variants are common, parental and segregation data are missing, and functional validation has not been performed; the observation is therefore hypothesis-generating rather than conclusive.

2. Case Presentation

The proband is a male member of a non-consanguineous family who was evaluated for developmental delay, intellectual disability, and strabismus. Precise ages at evaluation and at onset of symptoms were not available.

Family history revealed three affected individuals among six siblings (two females and one male) as well as an affected maternal male first cousin. This distribution is compatible with autosomal recessive inheritance, but the pattern cannot be confirmed genetically because only the proband was sequenced and no segregation studies were performed.

The clinical diagnosis was recorded as familial neurodevelopmental disorder with developmental delay and strabismus. No further clinical details—neuroimaging, seizure history, or systemic findings—were available.

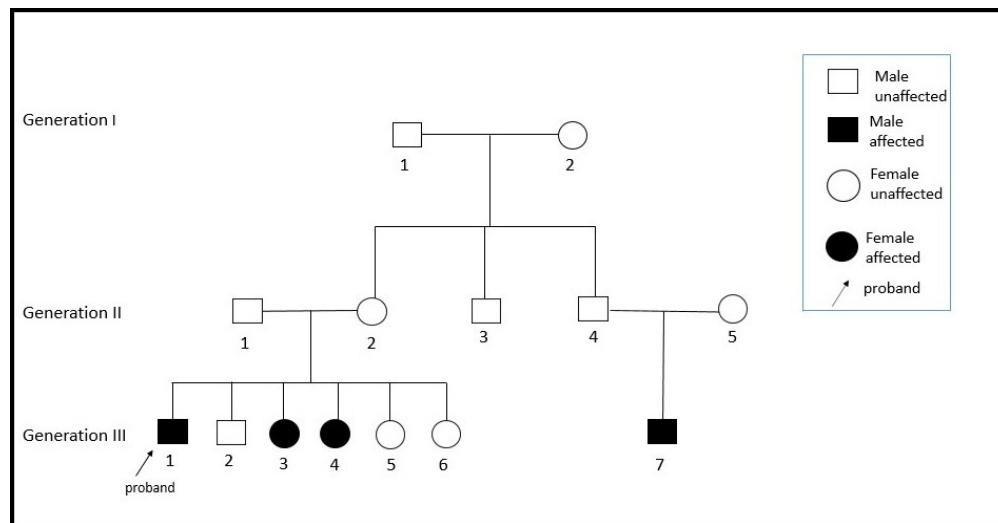


Figure 1. Pedigree of the family. The arrow marks the proband (individual 1 in Generation III). Filled symbols indicate affected individuals; open symbols indicate unaffected individuals. The distribution is consistent with possible autosomal recessive inheritance, although molecular confirmation is lacking.

3. Whole-Exome Sequencing and Variant Analysis

Whole-exome sequencing was performed on the proband. Variant prioritization followed a straightforward filtering pipeline. First, variants annotated as benign or likely benign were removed. Intronic variants were then

excluded. Genes previously reported in association with developmental delay, intellectual disability, or strabismus were inspected specifically; none harbored novel or convincingly pathogenic changes.

After these steps, two heterozygous missense variants in CX3CR1 remained: c.745G>A (p.Val249Ile; V249I; rs3732379) and c.839C>T (p.Thr280Met; T280M; rs3732378). Both have been characterized mainly as common functional polymorphisms or modest risk alleles rather than Mendelian pathogenic variants. They are known to be in strong linkage disequilibrium in most populations.

Because parental DNA and DNA from other relatives were unavailable, neither segregation nor phase (cis versus trans) could be established. Consequently, the mere presence of both heterozygous variants does not demonstrate pathogenicity of either allele alone or of their combination. On the basis of the clinical picture, the absence of alternative candidates, and the biological plausibility of CX3CR1, we regard the gene as a candidate rather than a proven disease gene.

Table 1. CX3CR1 variants identified by whole-exome sequencing

Gene	HGVS c.	Protein change	dbSNP	Type	Zygosity	Depth	AF
CX3CR1	c.745G>A	p.Val249Ile (V249I)	rs3732379	Missense	Heterozygous	77×	0.50
CX3CR1	c.839C>T	p.Thr280Met (T280M)	rs3732378	Missense	Heterozygous	74×	0.50

Both variants were detected in the heterozygous state and are listed as candidates rather than established pathogenic alleles. Parental testing, segregation analysis, and molecular phasing were unavailable.

4. Discussion

Whole-exome sequencing of the proband identified two heterozygous missense variants in CX3CR1—V249I and T280M—against a background of developmental delay, intellectual disability, and strabismus. No alternative candidates emerged from genes previously associated with the phenotype. These observations prompted us to consider CX3CR1 as a possible contributor.

CX3CR1 is the receptor for fractalkine (CX3CL1) and is expressed predominantly by microglia. The CX3CL1–CX3CR1 axis regulates microglial surveillance, synaptic pruning, and neuronal homeostasis. Mice lacking Cx3cr1 show impaired synaptic pruning, altered synaptic transmission, reduced functional connectivity, and behavioral abnormalities that overlap with neurodevelopmental phenotypes [3]. Animal data alone cannot prove that a human variant causes Mendelian disease, yet they supply a coherent biological framework.

Human genetic support is thinner but not absent. The rare p.Ala55Thr allele reported by Ishizuka et al. associated with schizophrenia and autism and impaired receptor signaling in functional assays [4]. That allele is distinct from the common polymorphisms found here, yet the study illustrates that CX3CR1 variation can influence neurodevelopmental risk.

V249I and T280M themselves have been investigated mainly as common functional polymorphisms. They alter ligand binding and cellular responses in vitro [5,6] and nearly always travel together because of strong linkage disequilibrium [7]. Clinical associations have been reported for amyotrophic lateral sclerosis survival, multiple-sclerosis course, and several inflammatory traits, but results vary by population and phenotype [8–10]. A large Japanese cohort also linked the haplotype to regional differences in cerebral arterial blood volume [11]. These findings confirm biological activity but fall well short of establishing Mendelian pathogenicity.

Interpretation in the present family is constrained by several missing pieces of evidence. Both variants were heterozygous; without parental genotypes or long-read phasing we cannot know whether they lie on the same chromosome or on opposite chromosomes. Segregation with the phenotype cannot be assessed because only the proband was sequenced. Standard frameworks for variant interpretation (population frequency, computational predictions, functional data, and segregation) [12] therefore remain incomplete.

The pedigree itself is nonetheless informative: three of six siblings and a maternal first cousin are affected, a pattern compatible with recessive inheritance. Molecular confirmation is simply unavailable.

Taken together, three observations keep CX3CR1 on the candidate list: (1) a familial neurodevelopmental phenotype without an alternative molecular explanation, (2) the well-documented role of CX3CR1 in neuron–

microglia signaling, and (3) prior human and experimental evidence that genetic variation in the gene can affect neurological phenotypes. At the same time, the specific V249I/T280M combination has never been shown to cause a Mendelian neurodevelopmental disorder.

Key limitations are the absence of segregation data, the unknown phase of the two variants, the lack of functional studies on the patient's genotype, and the limited clinical detail available for the family. Further work should include testing of affected and unaffected relatives, molecular phasing, cellular assays of receptor function, and the identification of additional unrelated individuals who carry rare or compound CX3CR1 variants and similar phenotypes.

In short, the present case raises the possibility that CX3CR1 contributes to a familial neurodevelopmental phenotype, but the evidence is insufficient to claim causality. The gene merits further investigation rather than immediate inclusion among confirmed Mendelian disease genes.

5. Conclusion

Whole-exome sequencing identified two heterozygous CX3CR1 polymorphisms, p.Val249Ile (V249I) and p.Thr280Met (T280M), in a patient with familial developmental delay, intellectual disability, and strabismus. Although these alleles have been studied mainly as common risk modifiers, their presence in this family, combined with the established biology of CX3CR1, justifies considering the gene a candidate. Segregation analysis, parental testing, phasing, functional validation, and additional independent cases are all required before a causal relationship can be asserted.

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